

Schematic V1.0

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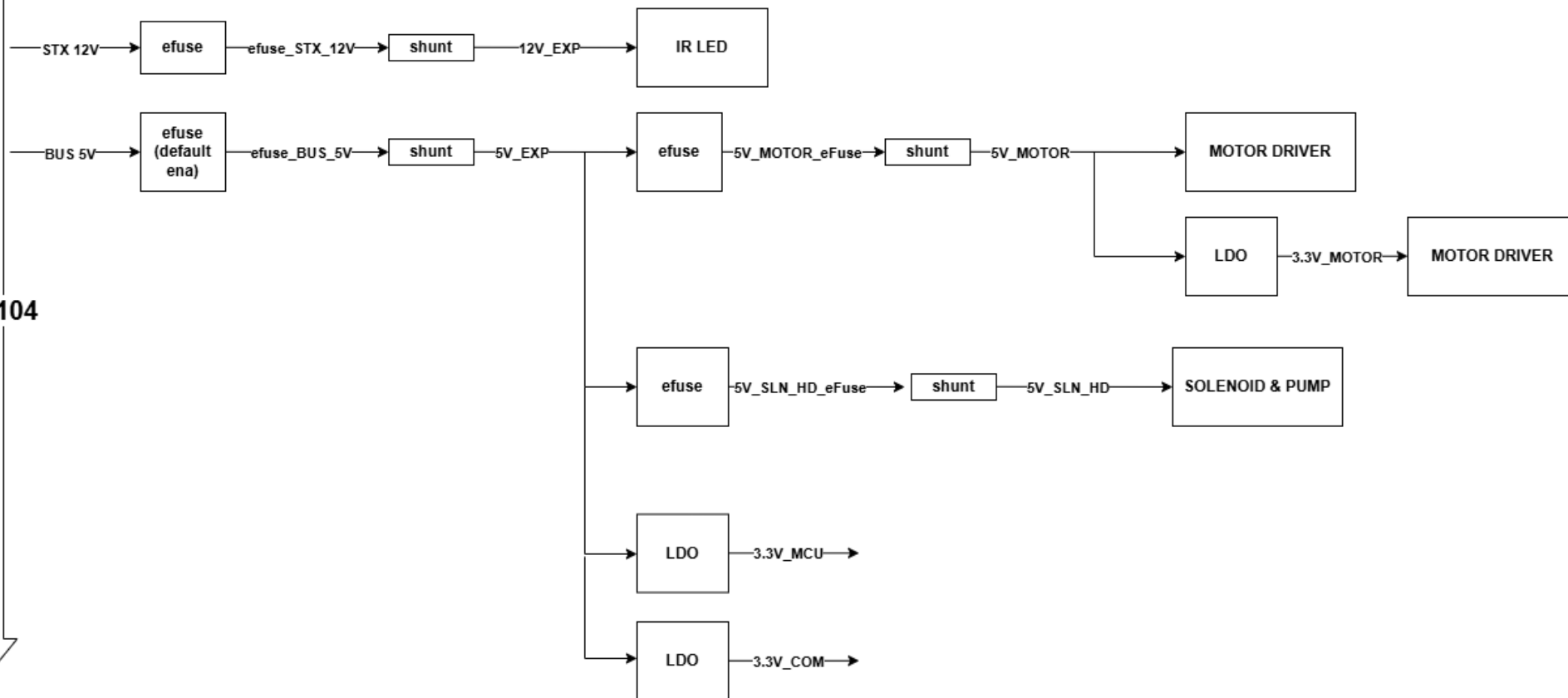
History

VERSION	DISCRIPTION
1.0	INITIAL PROJECT



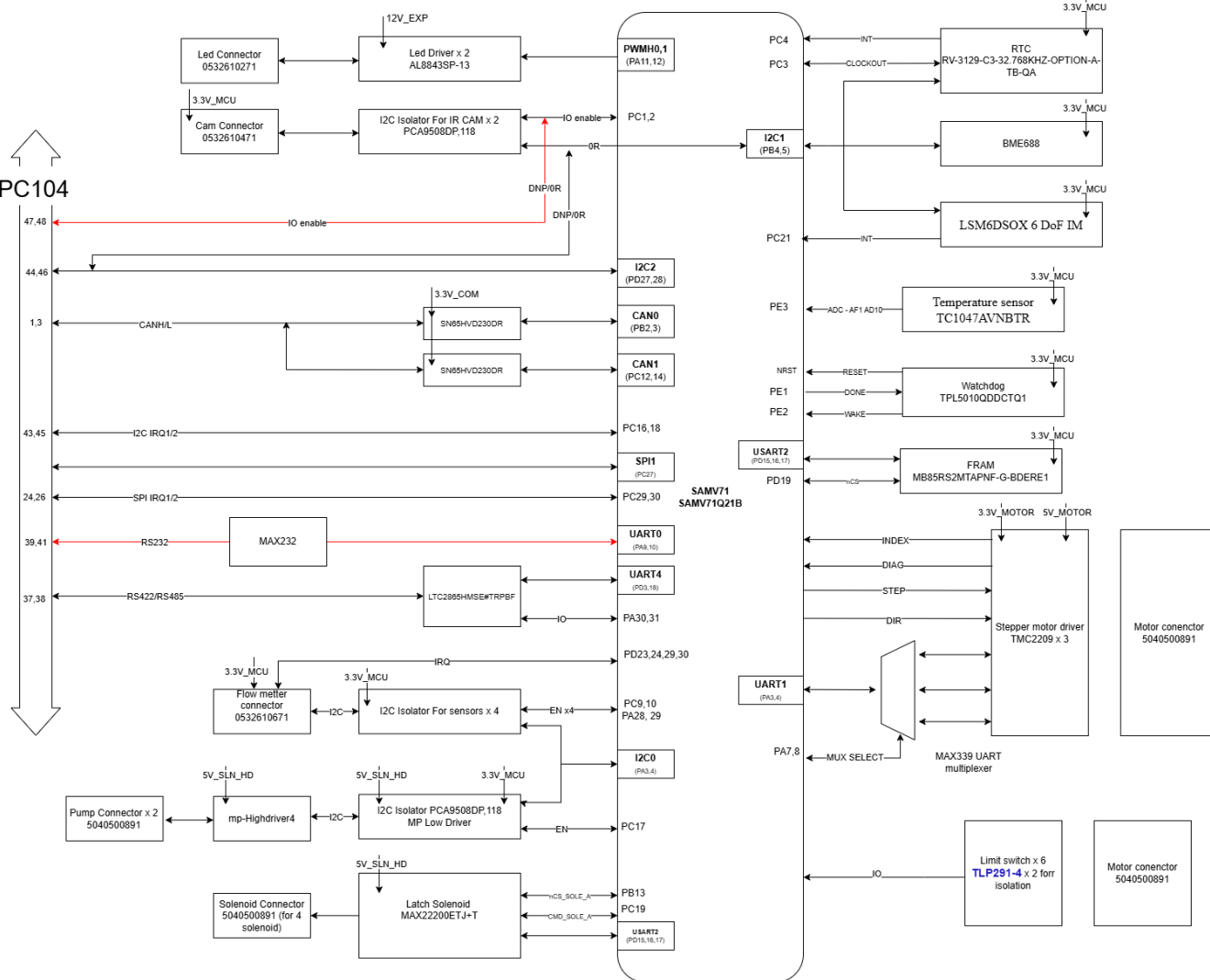
Project	BEE_DAL	Sub-system:	BEE-DAL_EXP_V1.0.PrjPcb
Design by:	Duy Nguyen	Sheet title: 01_INDEX.SchDoc	Sheet: 1 of 27
Checked by:	Bao Bui	Doc no: 1	Size: A3
Approved by:	Vu Pham	Revision 1.0.0	Date: 18-Aug-26

PC104



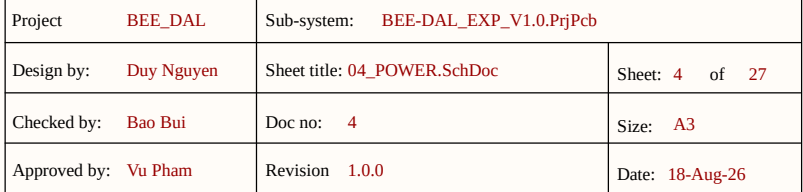
Project	BEE_DAL	Sub-system:	BEE-DAL_EXP_V1.0.PrjPcb
Design by:	Duy Nguyen	Sheet title: 02_POWER_TREE.SchDoc	Sheet: 2 of 27
Checked by:	Bao Bui	Doc no: 2	Size: A3
Approved by:	Vu Pham	Revision 1.0.0	Date: 18-Aug-26

PC104



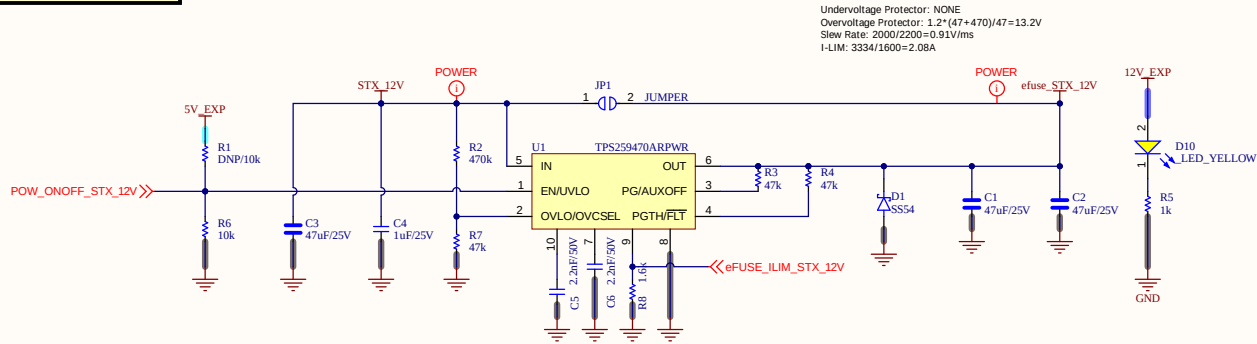
Project	BEE_DAL	Sub-system:	BEE-DAL_EXP_V1.0.PrtPcb
Design by:	Duy Nguyen	Sheet title:	03_BLOCK_DIAGRAM.SchDoc
Checked by:	Bao Bui	Doc no:	3
Approved by:	Vu Pham	Revision	1.0.0
		Sheet:	3 of 27
		Size:	A3
		Date:	18-Aug-26

LDO for driver (driver communication)

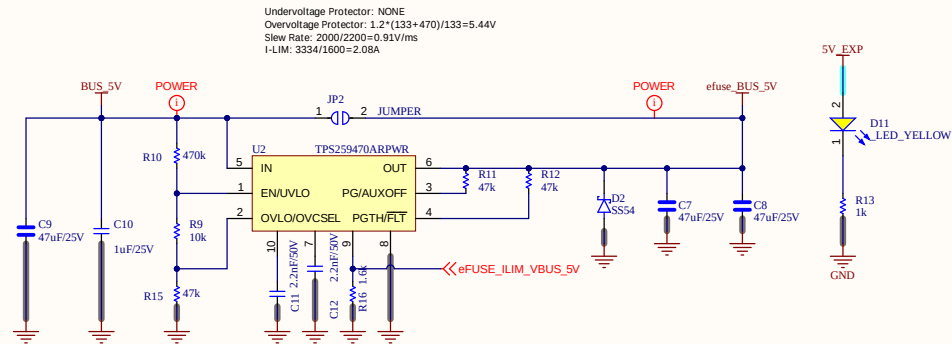


* 2 efuse dùng cho nguồn tổng là STX 12V và BUS 5V, trong đó STX 12V sẽ có thể được MCU on/off, còn BUS 5V thì mặc định on (do MCU lấy 5V từ đây)

EFUSE STX 12V

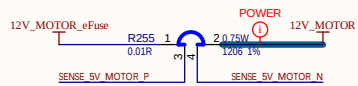
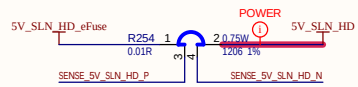
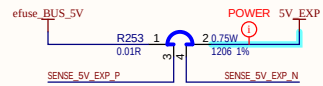
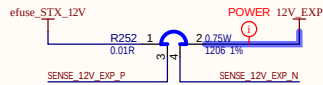
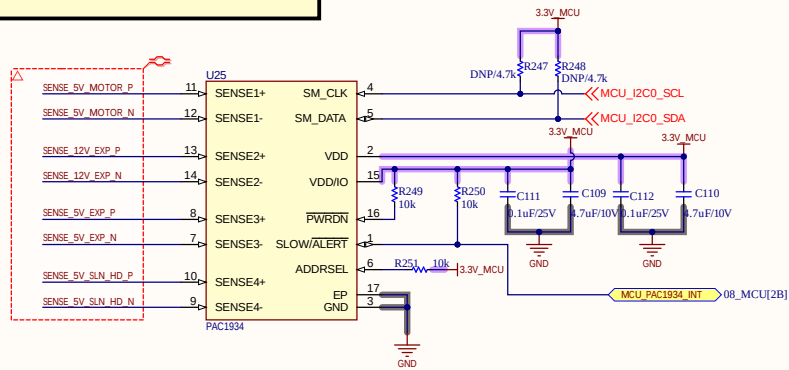


EFUSE BUS 5V



Project	BEE_DAL	Sub-system:	BEE-DAL_EXP_V1.0.PrjPcb
Design by:	Duy Nguyen	Sheet title:	05_POWER_EFUSE.SchDoc
Checked by:	Bao Bui	Doc no:	5
Approved by:	Vu Pham	Revision	1.0.0
		Sheet:	5 of 27
		Size:	A3
		Date:	18-Aug-26

POWER MEASUREMENT 12V AND 5V



Project	BEE_DAL	Sub-system:	BEE-DAL_EXP_V1.0.PrjPcb
Design by:	Duy Nguyen	Sheet title:	06_POWER_MEASUREMENT.SchDoc
Checked by:	Bao Bui	Doc no:	6
Approved by:	Vu Pham	Revision	1.0.0
		Size:	A3
		Date:	18-Aug-26

A

B

C

D

EFUSE for SLN & HD

Undervoltage Protector: NONE
 Overvoltage Protector: $1.2 \times (133 + 470) / 133 = 5.44V$
 Slew Rate: $2000 / 2200 = 0.91V/ms$
 I-LIM: $3334 / 1600 = 2.08A$

Component Values:

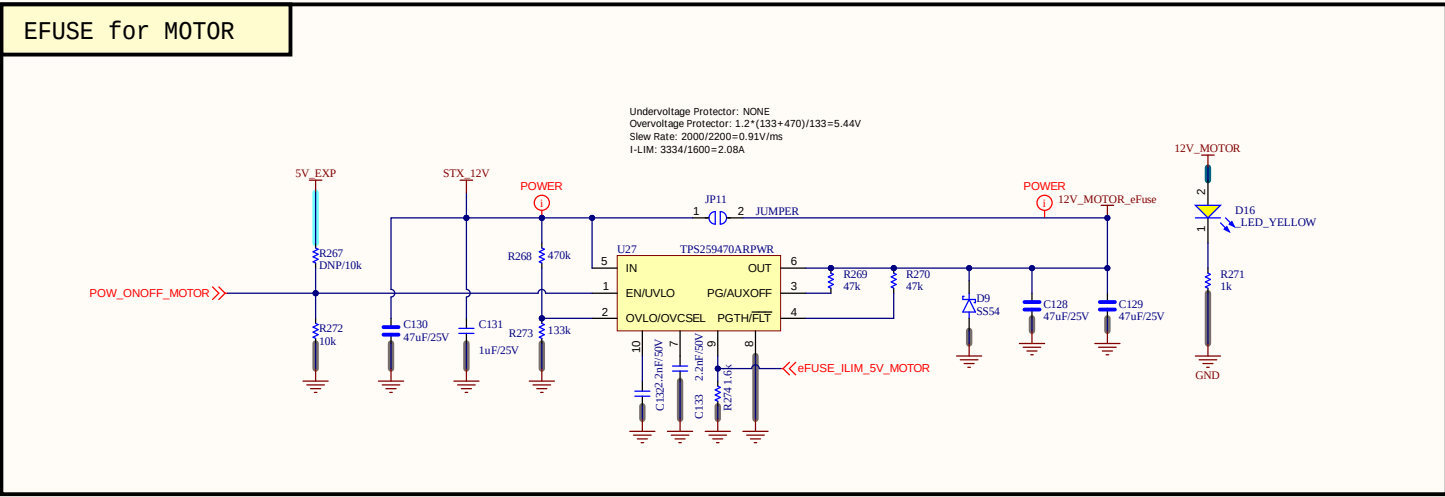
Component	Value
R259	DNP/10k
R264	10k
C124	47uF/25V
C125	1uF/25V
R260	470k
R265	133k
JP10	JUMPER
R261	47k
R262	47k
D8	SS54
C122	47uF/25V
C123	47uF/25V
R263	1k

Pin Connections:

- IN: 10
- EN/UVLO: 1
- PG/AUXOFF: 3
- OVLO/OVCSEL: 2
- PGTH/FLT: 4
- OUT: 6

Labels:

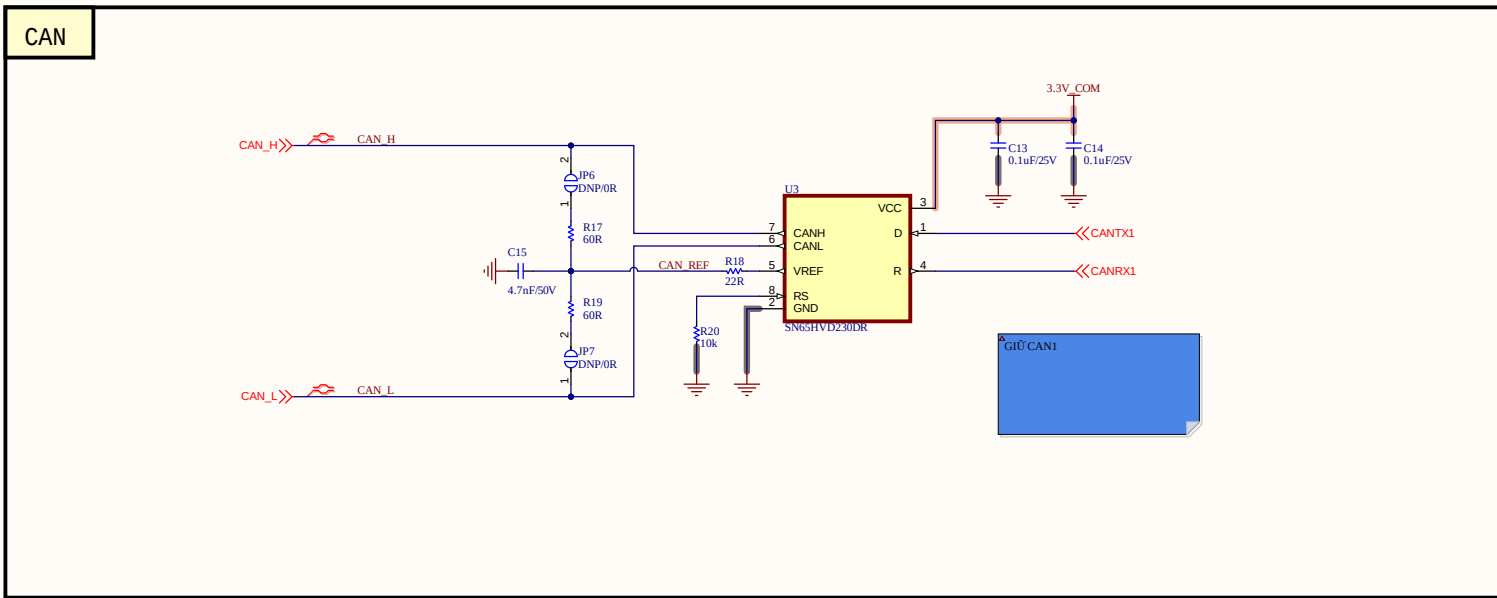
- 5V_EXP
- POWER
- 5V_SLN_HD
- POW_ONOFF_SLN_HD
- 5V_SLN_HD_eFuse
- D15_LED_YELLOW
- GND

[illegible]

Project	BEE_DAL	Sub-system: BEE-DAL_EXP_V1.0.PrjPcb	
Design by:	Duy Nguyen	Sheet title: 07_FUNCTION_EFUSE.SchDoc	Sheet: 7 of 27
Checked by:	Bao Bui	Doc no: 7	Size: A3
Approved by:	Vu Pham	Revision 1.0.0	Date: 18-Aug-26

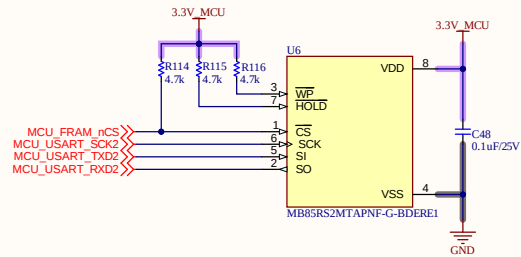


Symbol	Parameter	Conditions	Min.	Typ.	Max.	Unit
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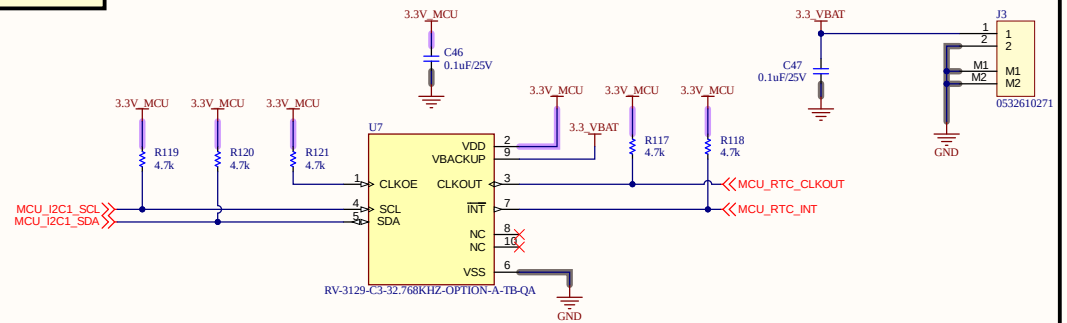


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Design by:	Duy Nguyen	Sheet title: 09_CAN.SchDoc	Sheet: 9 of 27
Checked by:	Bao Bui	Doc no: 9	Size: A3
Approved by:	Vu Pham	Revision 1.0.0	Date: 18-Aug-26

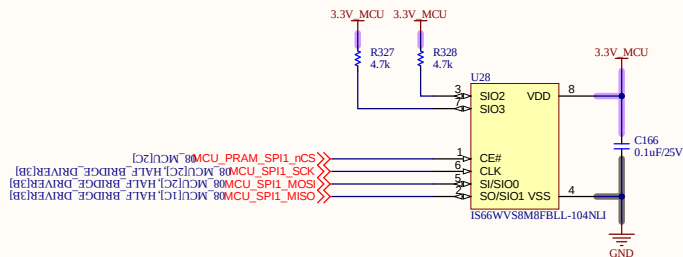
FRAM 2M



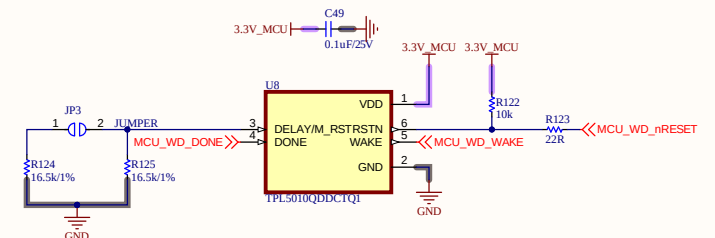
RTC



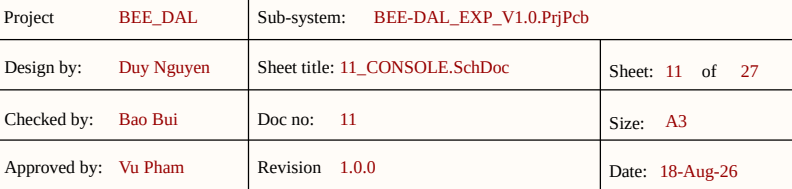
PRAM 64M

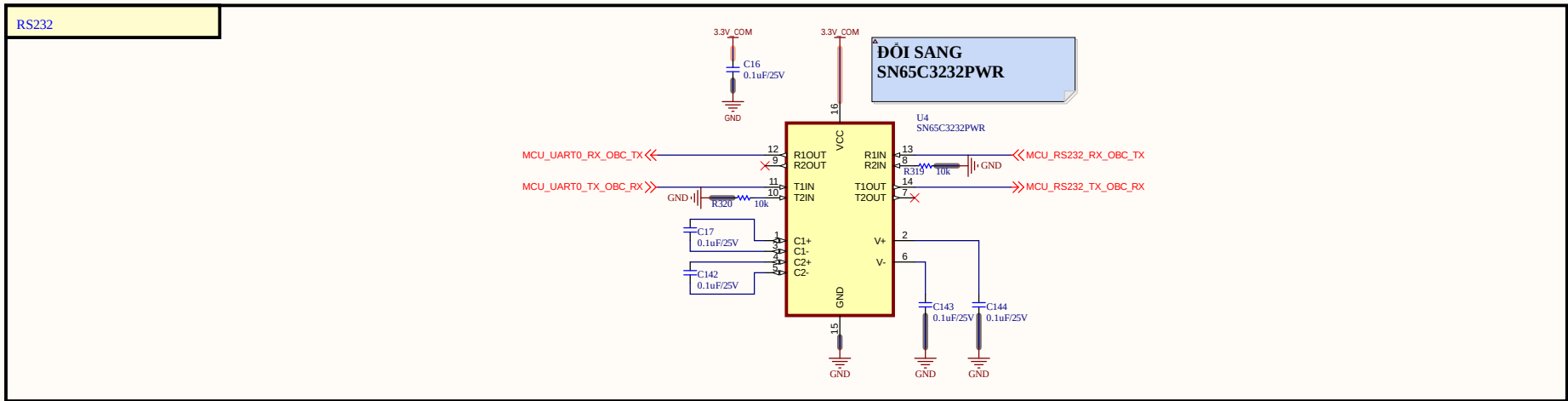
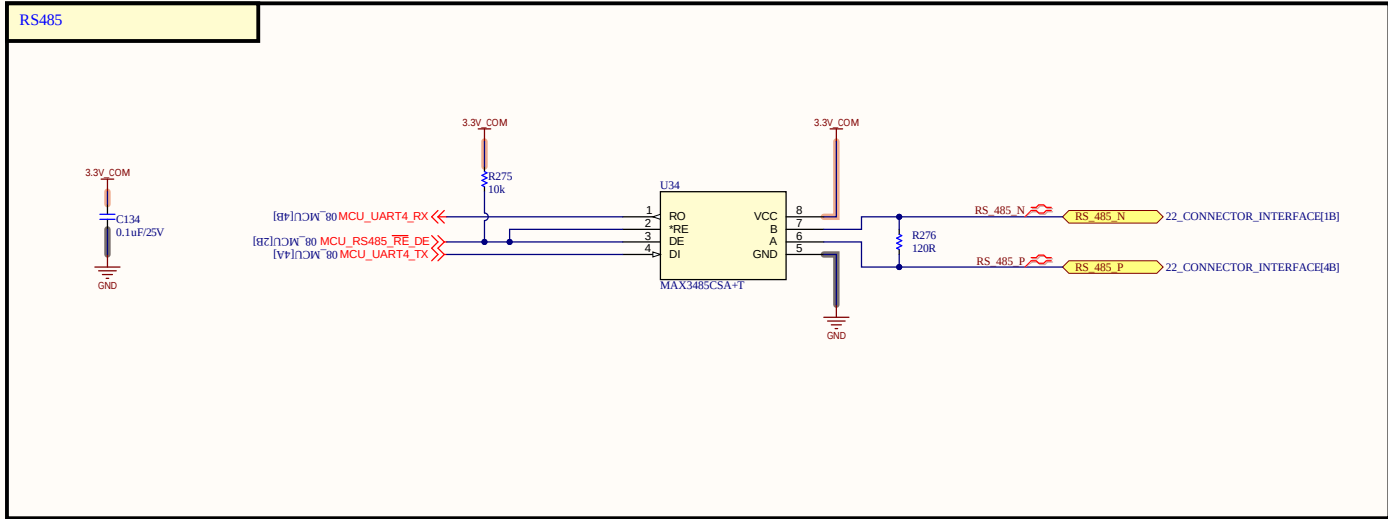


WATCH DOG



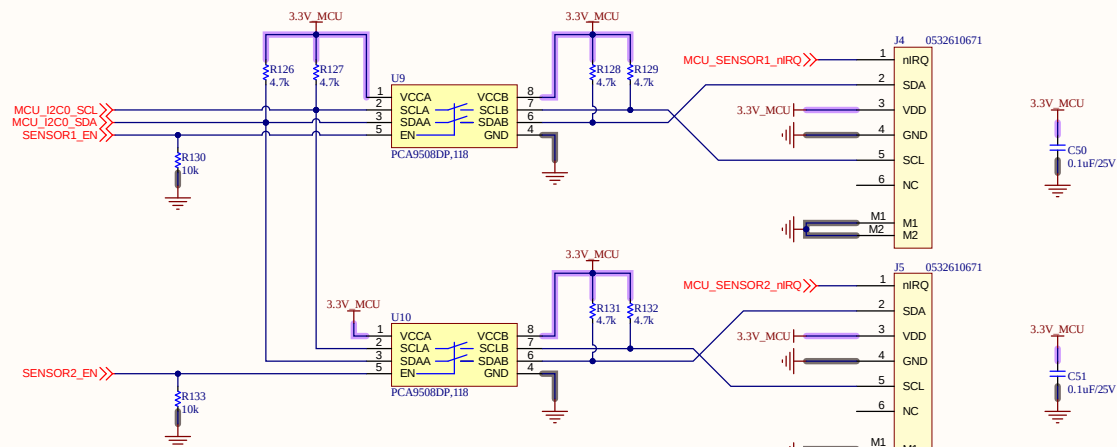
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Design by:	Duy Nguyen	Sheet title:	10_PRAM_RTC_WDT.SchDoc
Checked by:	Bao Bui	Doc no:	10
Approved by:	Vu Pham	Revision	1.0.0
		Sheet:	10 of 27
		Size:	A3
		Date:	18-Aug-26





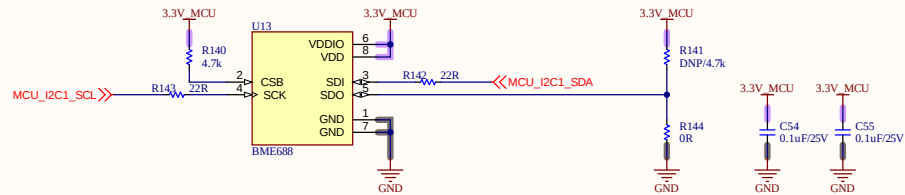
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Design by:	Duy Nguyen	Sheet title:	12_RS485_R232.SchDoc
Checked by:	Bao Bui	Doc no:	12
Approved by:	Vu Pham	Revision	1.0.0
		Sheet:	12 of 27
		Size:	A3
		Date:	18-Aug-26

I2C

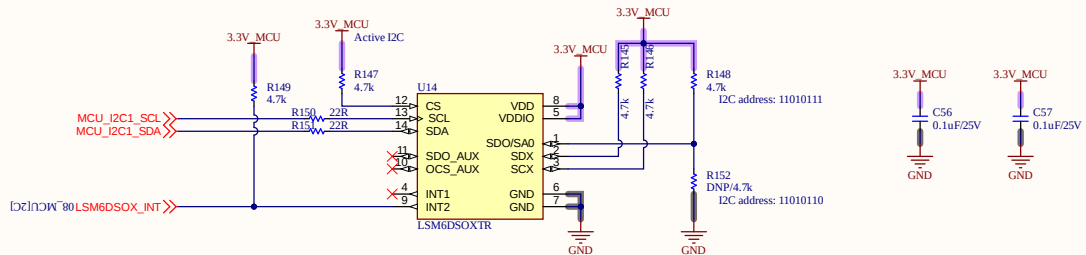


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Design by:	Duy Nguyen	Sheet title:	13_FLOW_Sensor.SchDoc
Checked by:	Bao Bui	Doc no:	13
Approved by:	Vu Pham	Revision	1.0.0
		Sheet:	13 of 27
		Size:	A3
		Date:	18-Aug-26

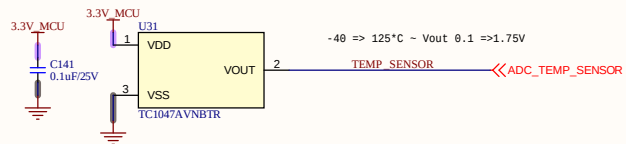
BME688



LSM6DSOX

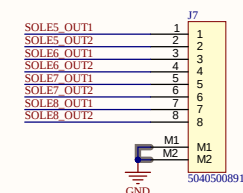
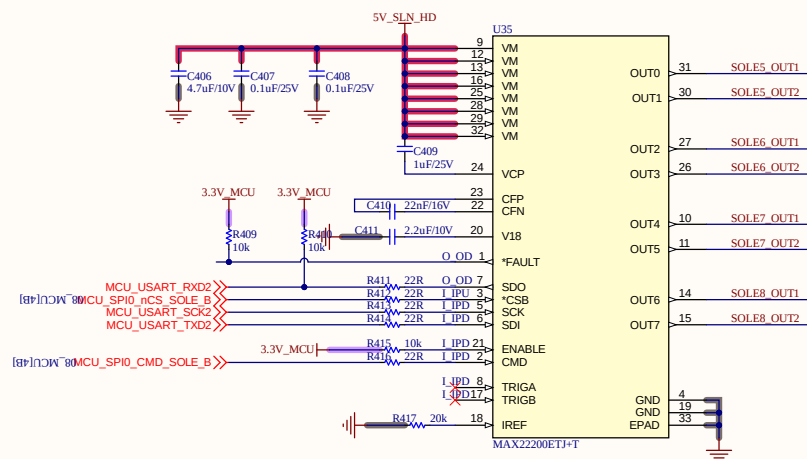
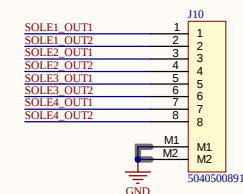
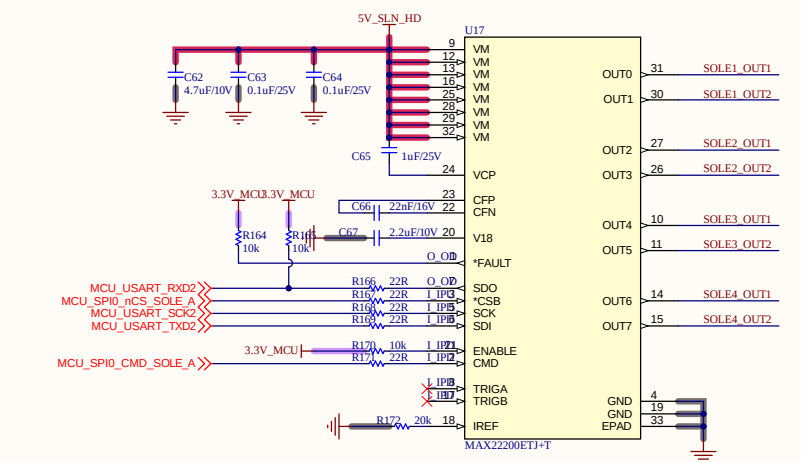


TEMPERATURE SENSOR

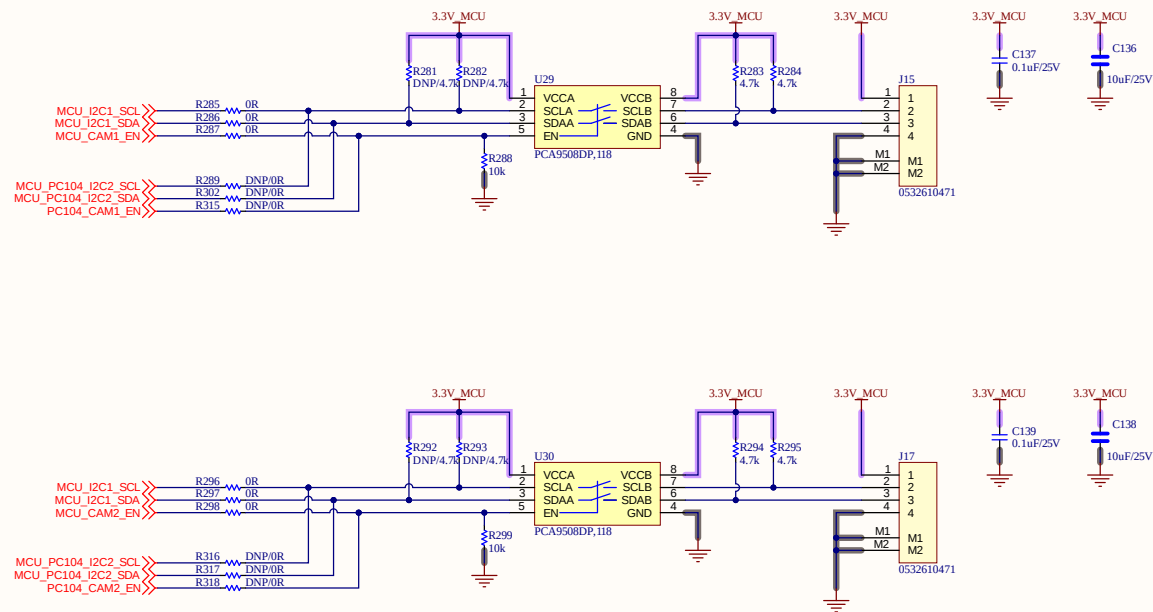


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Design by:	Duy Nguyen	Sheet title:	14_I2C_Sensor_OnBoard.SchDoc
Checked by:	Bao Bui	Doc no:	14
Approved by:	Vu Pham	Revision	1.0.0
		Sheet:	14 of 27
		Size:	A3
		Date:	18-Aug-26

SOLENOID

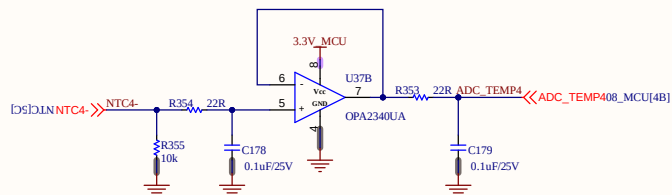
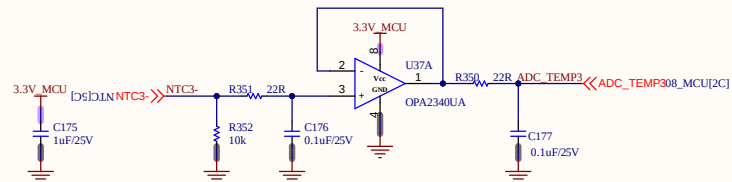
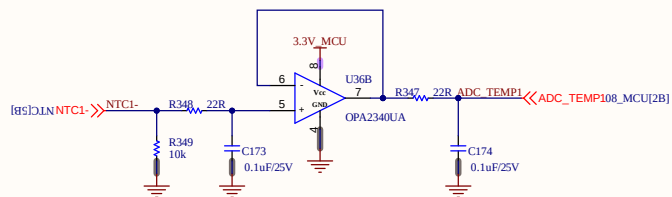
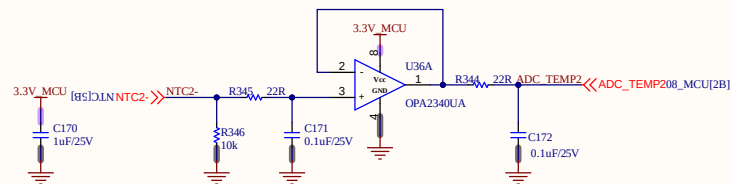


Project	BEE_DAL	Sub-system: BEE-DAL_EXP_V1.0.PrtPcb	
Design by:	Duy Nguyen	Sheet title: 16_SOLENOID_HB_1.SchDoc	Sheet: 16 of 27
Checked by:	Bao Bui	Doc no: 16	Size: A3
Approved by:	Vu Pham	Revision 1.0.0	Date: 18-Aug-26

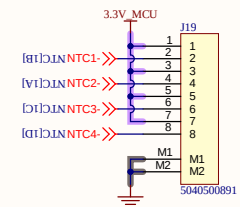


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Design by:	Duy Nguyen	Sheet title: 17_CAM_CONNECTOR.SchDoc	Sheet: 17 of 27
Checked by:	Bao Bui	Doc no: 17	Size: A3
Approved by:	Vu Pham	Revision 1.0.0	Date: 18-Aug-26

NTC

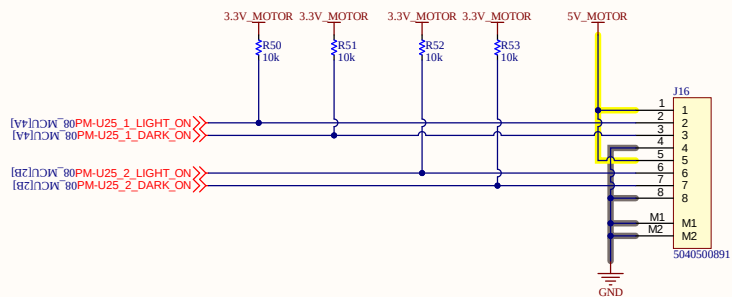


CONNECTOR FOR NTC

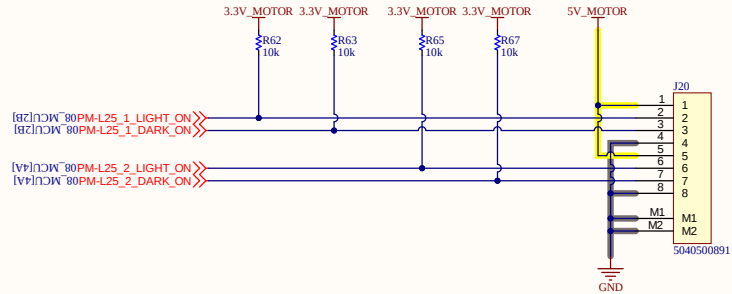


Project	BEE_DAL	Sub-system:	BEE-DAL_EXP_V1.0.PrjPcb
Design by:	Duy Nguyen	Sheet title:	NTC.SchDoc
Checked by:	Bao Bui	Doc no:	20
Approved by:	Vu Pham	Revision	1.0.0
		Sheet:	20 of 27
		Size:	A3
		Date:	18-Aug-26

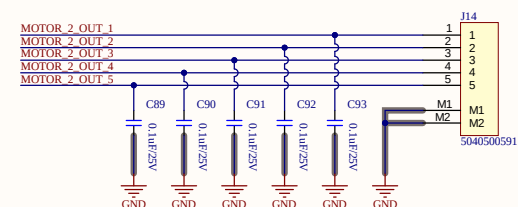
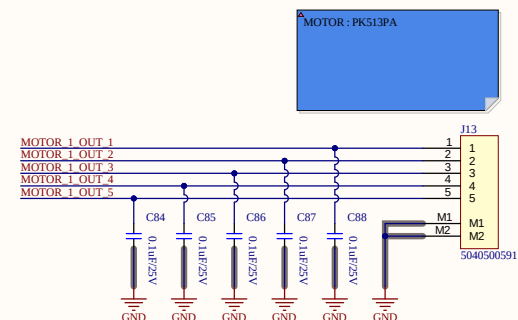
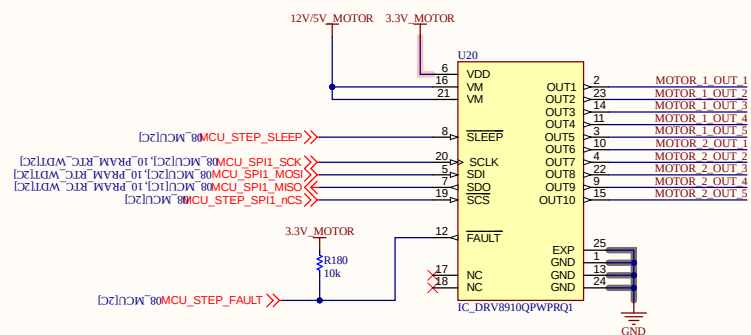
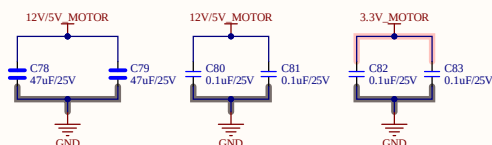
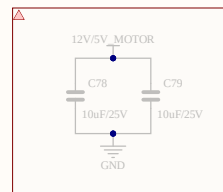
CONNECTOR for PM-U25



CONNECTOR for PM-L25



Project	BEE_DAL	Sub-system:	BEE-DAL_EXP_V1.0.PrjPcb
Design by:	Duy Nguyen	Sheet title:	LIMIT_SENSORS.SchDoc
Checked by:	Bao Bui	Doc no:	21
Approved by:	Vu Pham	Revision	1.0.0
		Sheet:	21 of 27
		Size:	A3
		Date:	18-Aug-26



Project	BEE_DAL	Sub-system: BEE-DAL_EXP_V1.0.PrjPcb		
Design by:	Duy Nguyen	Sheet title: HALF_BRIDGE_DRIVER.SchDoc	Sheet: 22	of 27
Checked by:	Bao Bui	Doc no: 22	Size: A3	
Approved by:	Vu Pham	Revision 1.0.0	Date: 18-Aug-26	

UART MUX

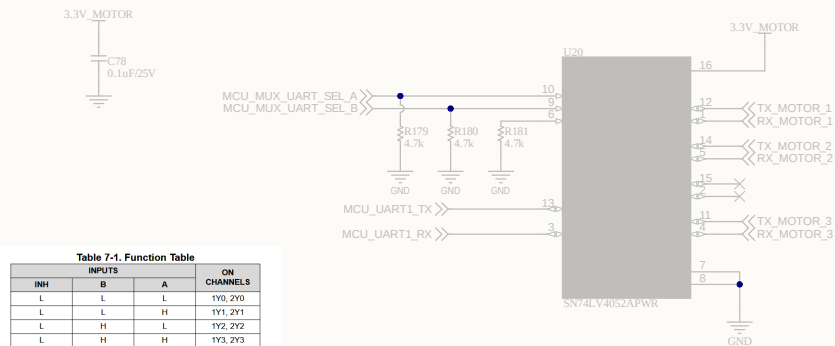
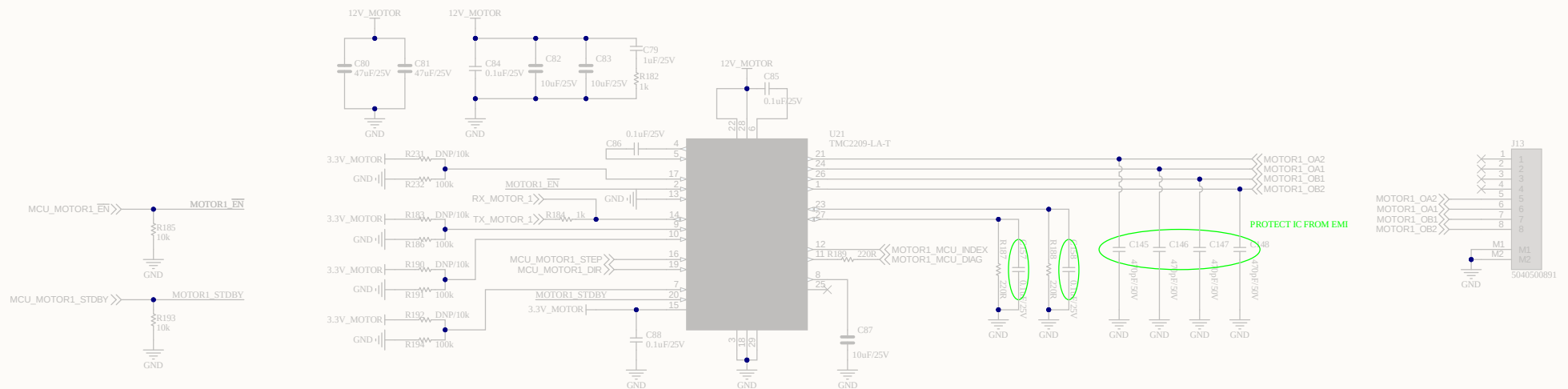


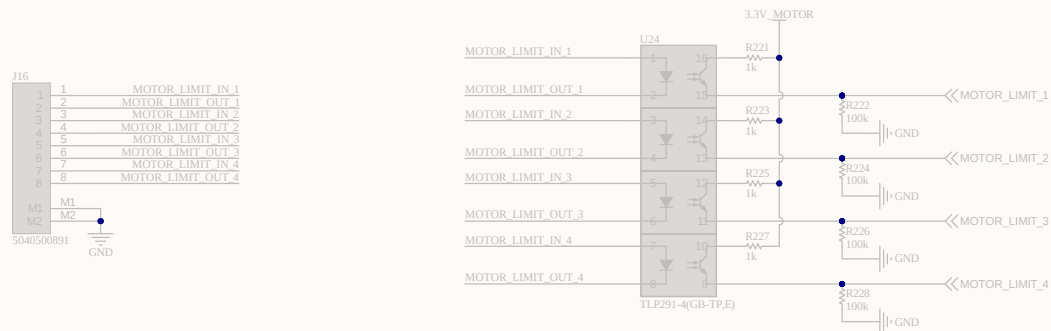
Table 7-1. Function Table

INPUTS			ON CHANNELS
INH	B	A	
L	L	L	1Y0, 2Y0
L	L	H	1Y1, 2Y1
L	H	L	1Y2, 2Y2
L	H	H	1Y3, 2Y3
H	X	X	None



Project	BEE_DAL	Sub-system:	BEE-DAL_EXP_V1.0.PrjPcb
Design by:	Duy Nguyen	Sheet title:	19_STEP_MOTOR_DRIVER_1.SchDoc
Checked by:	Bao Bui	Doc no:	23
Approved by:	Vu Pham	Revision	1.0.0
		Date:	18-Aug-26

STEP LIMIT



Project	BEE_DAL	Sub-system:	BEE-DAL_EXP_V1.0.PrjPcb
Design by:	Duy Nguyen	Sheet title:	21_STEP_MOTOR_ISOLATOR_INOUT_SchDoc
Checked by:	Bao Bui	Doc no:	25
Approved by:	Vu Pham	Revision	1.0.0
		Size:	A3
		Date:	18-Aug-26

